Surya J

APP & WEB DEVELOPER

ANDROID · FULL-STACK · AI SYSTEMS

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SUMMARY

App and web developer building production Android and full-stack systems. Software developer at Kosal Tech Solutions, with a contract engagement at Manju Global — delivering Kotlin/Compose mobile apps, Next.js platforms, and the Go and Python services behind them. B.E. Computer Science & Engineering, 2022–2026. Focused on AI-powered applications and architecture that stays readable for whoever maintains it next.

TECHNICAL SKILLS

LANGUAGES	Python, JavaScript, TypeScript, Kotlin, Dart, Go, SQL
MOBILE	Android (Kotlin, Jetpack Compose, MVVM), Material 3, Flutter
WEB	React, Next.js, Node.js, HTML, CSS, Tailwind CSS
BACKEND	FastAPI, Express.js, Convex, Firebase, REST APIs, WebSockets
DATA	PostgreSQL, PostGIS, TimescaleDB, MongoDB, Redis, SQLite, Firestore
AI / ML	Machine Learning, Generative AI, Prompt Engineering, scikit-learn, LLM APIs
TOOLING	Git, GitHub, Docker, Gradle, VS Code, Android Studio
DEPLOYMENT	Vercel, Render, Google Play Store

EXPERIENCE

Software Developer — CURRENT

Kosal Tech Solutions

- Build internal tooling and client-facing web systems across the company's delivery work.
- Developed Kosal-Link, a four-tool scraper suite — static downloader, browser-based scraper for JS-heavy pages, structure extractor, and a CLI that renders a live landing page, detects section boundaries, and exports selector-scoped components as a reusable blueprint.
- Built and shipped client landing pages and marketing sites end to end.

App & Web Developer · Contract — CONTRACT

Manju Global

- Mconnect — Android client for the Manju Groups project-management system: OTP authentication, HR, attendance, leave and chat. Kotlin + MVVM + Retrofit, AES256-GCM encrypted session storage, and a semantic colour-token design system driving light and dark from one source.
- Manju PMS platform — Next.js + Convex operations platform covering land procurement, legal and feasibility gating, channel-partner management, Excel/PDF reporting and an AI sales assistant.
- Airix Geo Tracking — Go service for location ingestion, live tracking, trip sessions and tamper monitoring on PostgreSQL/PostGIS + TimescaleDB, with Redis for live state and idempotency.
- Telecaller Core — extracted call operations into a standalone Go service: dialer webhooks, incoming/follow-up/outbound lanes and the outcome funnel, with retries, callbacks, rollups and outbox delivery running as durable Temporal workflows while the hot path stays one Postgres transaction.
- Travel Desk — Next.js trip allocation and fleet billing: token-scoped driver pages delivered over WhatsApp with no driver login, OCR on uploaded odometer evidence, and explicit extra-km, cancellation and offline-completion states.
- Airix iOS — SwiftUI counterpart to Mconnect on the same production backend, built to a written parity roadmap so behaviour matches the Android client while the interface stays native.
- WhatsApp platform — multi-tenant Next.js + Convex product over the WhatsApp Business Cloud API: templates, campaigns, shared inbox, idempotent sends and a server-to-server API other products use to send their own notifications.

MERN Stack Developer Intern — JAN–FEB 2025

VEI Technologies

- Developed full-stack modules using MongoDB, Express.js, React and Node.js.
- Built and integrated REST APIs connecting frontend components with backend services across agile sprints.

Web Development Intern — JAN 2025

NoviTech

- Designed responsive web interfaces in HTML, CSS and JavaScript, integrating backend services and optimising page performance across multiple modules.

PROJECTS

Mentorix — AI Career Intelligence System

`Python · FastAPI · RandomForest · SQLite · Vercel / Render` `github.com/D-A-R-X/mentorix-ai`

AI engine analysing student behaviour patterns to predict career instability and recommend personalised paths. Modular FastAPI REST services with a RandomForest model behind an explainable AI layer that returns reasoning, not just scores.

Darx Dialer — Default Phone App with Cloud Recording

`Kotlin · Jetpack Compose · InCallService · Convex`

Full default-dialer replacement implementing InCallService and the DIALER role — lock-screen incoming UI, Bluetooth routing, DTMF and hold. Per-call recording is consent-first: on-screen indicator, persistent notification and a user toggle, never covert.

Brownie Care — AI Companion Health App

`Kotlin · Compose Canvas · Groq LLM · Convex · WorkManager`

Health companion built around a character drawn entirely in Compose Canvas with spring animations — no image assets. DataStore is the source of truth with Convex as a best-effort cloud mirror, so the app works fully offline.

Megatron — Modular Adaptive AI Control System

`Python · FastAPI · JavaScript` `github.com/D-A-R-X/Megatron`

Modular AI controller with a central reasoning layer that analyses intent and delegates to specialised modules — new capabilities added without rewriting core logic.

Smart Campus — Academic Resource Management App

`Flutter · Firebase Auth · Firestore · Firebase Storage` `Play Store – com.schrodingerlab.curiosity`

Cross-platform app delivering centralised academic resources with offline download support. Deployed to the Google Play Store.

Sentinel X — Trading Signal Verification & Paper Trading

`Python · Telegram API · Zerodha Kite Connect · Pine Script · SQLite`

Records every call a signal channel makes and logs what would have happened, measuring true win rate without placing a real trade. An Indian-market build adds a virtual broker on Kite Connect with expiry and rollover rules and hard risk caps; live trading stays gated off.

Drive Track Mate — Intelligent Driving Monitoring

`Android · Dash Camera Integration · Event-Based Recording` `github.com/D-A-R-X/drive-track-mate`

Event-aware recording that auto-tags sudden braking and abrupt manoeuvres with timestamps and metadata, replacing unstructured continuous footage with retrievable evidence.

EDUCATION

B.E. Computer Science & Engineering

`2022 – 2026`

Dhanalakshmi Srinivasan College of Engineering

`Coimbatore, Tamil Nadu` `CGPA 7.43 / 10`

Coursework in data structures, databases, computer networks, operating systems and machine learning, alongside production Android and full-stack work throughout the degree.

CERTIFICATIONS

OCI 2023 Certified Foundations Associate · Oracle University	`Issued Jun 2024`
AI for Students: Build Your Own Generative AI Model · NxtWave	`Sep 2024`
Fundamentals of Digital Marketing · Google Digital Unlocked	`Jan 2021`

RECOGNITION

- Published 'AI in Retail Marketing' — PSG College Conference.
- MSME Hackathon — Rathinam Institute of Technology.
- Presented at 3+ national hackathons.
- MERN Stack Workshop — CIT Coimbatore, Feb 2025.

ALSO

Cruza — co-founder of an independent software studio building intelligent systems, est. 2025. Flagship product: Mentorix, a live career-intelligence system.